Paper Presentation

Title of the Paper

University of Information Technology

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Outline

1. Introduction
2. New strageries
3. Experiment
4. The effects of the first strategy

Background

Ant algorithms are easy to implement and cover wide range of applications, but their performance dramatically decreases in dealing with large-scale problems

ACO drawbacks

- ACO algorithms need (n2) space complexity to hold pheromone values in main memory.
- Each ant completes its solution in O(n2) time so the total time complexity becomes O(n2 x number of iterations) that is too high for applying the ACO to large instances.
- In recent years, local searches have been utilized in ACO algorithms to increase quality of solutions

First strategy

Using pheromone information in local search

- use pheromone information in calculating candidates-sets that used by local search
- initialized using nearest-neighbors

Algorithm 2. Candidates-set updating algorithm.
for each n in instance
$best = I$ where $\forall J \neq n$ then $\tau[n][J] \leq \tau[n][I]$ //variable *best* is argmax { $\tau[n][0]$, $\tau[n][1]$, ... $\tau[n][size - 1]$}
Remove the last node of candidates-set of node n.
Insert $best$ into the first place of candidates-set of node n.
Remove the last node of candidates-set of node n.
Insert *successor of* n in best tour into the first of candidates-set of node n.
end for

- node in the last place of candidates-set of n should be removed (structure of the candidates-set is similar a queue) and repetition is not allowed in candidates-sets.

Second strategy

Second strategy, new representation for pheromone values

Majority of pheromone values are initiated 0 and remained fix until the end of algorithm so it can be easily concluded that we are faced with sparse matrix

$$pheromones(i,j) = \begin{cases} \tau_0 \left(= \dfrac{1}{cost\ of\ the\ best\ solution\ in\ initial\ phase} \right), \\ corresponding\ value\ of\ node\ j\ ini's\ list, \end{cases}$$

Third strategy

ACO speedup

Algorithm 3. Usual algorithm to speed up transition rule.
Instructions for Ant_k, to select next node of node r:

1) Probe Eq. (4) for candidates of r at first and if choosing them is impossible go to next step.
2) Probe Eq. (4) for all not visited nodes.

Heuristically

item If a node, say j, has been selected as the next node of i in the previous steps so it has high chance to be selected as the next node of i in the current step.

To select next node of i, algorithm should not only probe candidate list of node i but also probe all nodes in corresponding list of node i in pheromone sparse matrix

Algorithm 4. Proposed algorithm to speed up transition rule.
Instructions for Ant_k, to select next node of node r:

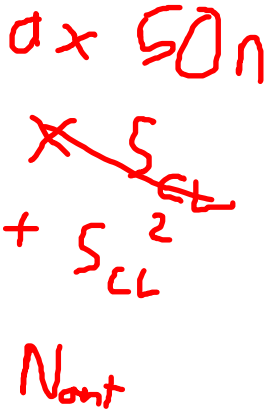
Hit domain:{
1) Probe Eq. (4) for candidates of *r* at first and if choosing of them are impossible, go to next step.
2) Probe Eq. (4) for all nodes in corresponding list of node *r* in pheromone spars matrix and if choosing of them are impossible, go to the next step.

Fault domain:{
3) Probe Eq. (4) for all not visited nodes.

Pseudo

Algorithm 5. Suggested ACO (ESACO) to solve large TSP instance.

```
1-  Apply initialization;
2-  for step = 1 to Max-Iterations
3-      for i = 1 to number-of-nodes
4-          for k = 1 to number-of-ants
5-              ant_k selects next edge according to Algorithm 4;
6-              ant_k update corresponding pheromone value of recent selected move by (3);
7-              if i = number-of-nodes
8-                  ant_k improve its tour by 2-opt local search;
9-              end if
10-         end for
11-     end for
12-     update global best tour;
10-     update pheromone globally belong to the global best tour;
11-     update candidates-sets by Algorithm 2;
12- end for
```





Experiments

Second strategy analysis: number of inserted items into spars matrix was less than 50 x n.

Third strategy analysis: hit rate values were greater than 0.97

$$hit\ domain\ rate\ =\ \frac{number\ of\ calling\ next\ cityfunction\ in\ hit\ domoin}{number\ calling\ second\ selection\ function}$$

Compared ESACO with TSIACO, MMAS, ACS by Wilcoxon signed ranks

Compared ESACO in large-scale: more stable and faster

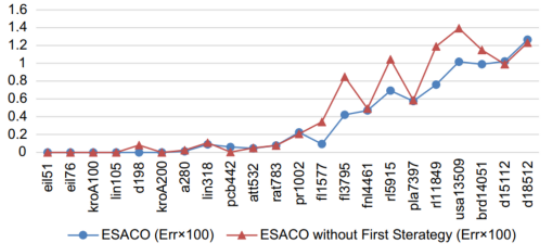


Fig. 7. Err of average cost: In performed experiment, for majority of instances, ESACO has better results than the case without using first strategy.

